SE LAB3

SRN - PES2UG24CS115
NAME - BHEEMA VARSHINI

Webhook Ingestion & Retry Mechanism Hub Architecture Choice

For the Webhook Ingestion & Retry Mechanism Hub, we decided to choose the Microservices Architecture.

Reason 1: The system performs multiple functions such as webhooks ingestion, webhook delivery, delivery attempts logging, retries execution, and failed webhooks storing. The fact that each function can be executed separately by one particular service makes this system more manageable.

Reason 2: The system must handle up to 2,500 requests per second and provide the reliability equal to 99.999%. By using the microservices architecture, we can scale only those parts of the system, which are needed, and at the same time prevent any problem in one service from spreading to other services.

Security Advantages: The access to the data of the webhooks can be restricted to authorized persons and services. The access of the operators to the stored data can be logged.

Performance Advantages: The services, which are responsible for webhooks receiving and processing, can be scaled based on the load of requests. This approach is especially useful if there is a high traffic which needs to be processed but not all services need scaling.

